



Lakshya JEE 2.0 (2024)

Indefinite Integral

Assignment

1. $\int \frac{\sin x}{\sin(x-\alpha)} dx =$
 - (1) $x \cos \alpha - \sin \alpha \log \sin(x-\alpha) + c$
 - (2) $x \cos \alpha + \sin \alpha \log \sin(x-\alpha) + c$
 - (3) $x \sin \alpha - \sin \alpha \log \sin(x-\alpha) + c$
 - (4) None of these
2. $\int \frac{\cos x - 1}{\cos x + 1} dx =$
 - (1) $2 \tan \frac{x}{2} - x + c$
 - (2) $\frac{1}{2} \tan \frac{x}{2} - x + c$
 - (3) $x - \frac{1}{2} \tan \frac{x}{2} + c$
 - (4) $x - 2 \tan \frac{x}{2} + c$
3. $\int \frac{dx}{1 - \sin x} =$
 - (1) $x + \cos x + c$
 - (2) $1 + \sin x + c$
 - (3) $\sec x - \tan x + c$
 - (4) $\sec x + \tan x + c$
4. If $\int (\sin 2x - \cos 2x) dx = \frac{1}{\sqrt{2}} \sin(2x - a) + b$, then
 - (1) $a = \frac{\pi}{4}, b = 0$
 - (2) $a = -\frac{\pi}{4}, b = 0$
 - (3) $a = \frac{5\pi}{4}, b = \text{any constant}$
 - (4) $a = -\frac{5\pi}{4}, b = \text{any constant}$
5. $\int \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right) dx =$
 - (1) $-e^x + c$
 - (2) $e^x + c$
 - (3) $e^{-x} + c$
 - (4) $-e^{-x} + c$
6. $\int \frac{\cot x \tan x}{\sec^2 x - 1} dx =$
 - (1) $\cot x - x + c$
 - (2) $-\cot x + x + c$
 - (3) $\cot x + x + c$
 - (4) $-\cot x - x + c$
7. $\int (\sec x + \tan x)^2 dx =$
 - (1) $2(\sec x + \tan x) - x + c$
 - (2) $1/3(\sec x + \tan x)^3 + c$
 - (3) $\sec x(\sec x + \tan x) + c$
 - (4) $2(\sec x + \tan x) + c$
8. $\int x^{51} (\tan^{-1} x + \cot^{-1} x) dx =$
 - (1) $\frac{x^{52}}{52} (\tan^{-1} x + \cot^{-1} x) + c$
 - (2) $\frac{x^{52}}{52} (\tan^{-1} x - \cot^{-1} x) + c$
 - (3) $\frac{\pi x^{52}}{104} + \frac{\pi}{2} + c$
 - (4) $\frac{x^{52}}{52} + \frac{\pi}{2} + c$
9. $\int 5 \sin x dx =$
 - (1) $5 \cos x + c$
 - (2) $-5 \cos x + c$
 - (3) $5 \sin x + c$
 - (4) $-5 \sin x + c$
10. $\int \frac{\tan x}{\sec x + \tan x} dx =$
 - (1) $\sec x + \tan x - x + c$
 - (2) $\sec x - \tan x + x + c$
 - (3) $\sec x + \tan x + x + c$
 - (4) $-\sec x - \tan x + x + c$
11. $\int \frac{dx}{\sin^2 x \cos^2 x} =$
 - (1) $\tan x + \cot x + c$
 - (2) $\cot x - \tan x + c$
 - (3) $\tan x - \cot x + c$
 - (4) None of these
12. $\int \left(x + \frac{1}{x} \right)^3 dx =$
 - (1) $\frac{1}{4} \left(x + \frac{1}{x} \right)^4 + c$
 - (2) $\frac{x^4}{4} + \frac{3x^2}{2} + 3 \log x - \frac{1}{2x^2} + c$
 - (3) $\frac{x^4}{4} + \frac{3x^2}{2} + 3 \log x + \frac{1}{x^2} + c$
 - (4) None of these

13. $\int \sqrt{1 + \sin \frac{x}{2}} dx =$

- (1) $\frac{1}{4} \left(\cos \frac{x}{4} - \sin \frac{x}{4} \right) + c$
- (2) $4 \left(\cos \frac{x}{4} - \sin \frac{x}{4} \right) + c$
- (3) $4 \left(\sin \frac{x}{4} - \cos \frac{x}{4} \right) + c$
- (4) $4 \left(\sin \frac{x}{4} + \cos \frac{x}{4} \right) + c$

14. $\int (\sin^{-1} x + \cos^{-1} x) dx =$

- (1) $\frac{1}{2} \pi x + c$
- (2) $x(\sin^{-1} x - \cos^{-1} x) + c$
- (3) $x(\cos^{-1} x + \sin^{-1} x) + c$
- (4) $\frac{\pi}{2} + x + c$

15. $\int \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}} dx =$

- (1) $\sin x + c$
- (2) $\cos x + c$
- (3) $x + c$
- (4) $x^2 + c$

16. $\int \frac{x-1}{(x+1)^2} dx =$

- (1) $\log(x+1) + \frac{2}{x+1} + c$
- (2) $\log(x+1) - \frac{2}{x+1} + c$
- (3) $\frac{2}{x+1} - \log(x+1) + c$
- (4) None of these

17. $\int \frac{\operatorname{cosec} \theta - \cot \theta}{\operatorname{cosec} \theta + \cot \theta} d\theta =$

- (1) $2 \operatorname{cosec} \theta - 2 \cot \theta - \theta + c$
- (2) $2 \operatorname{cosec} \theta - 2 \cot \theta + \theta + c$
- (3) $2 \operatorname{cosec} \theta + 2 \cot \theta - \theta + c$
- (4) None of these

18. $\int (1 + 2x + 3x^2 + 4x^3 + \dots) dx =$

- (1) $(1+x)^{-1} + c$
- (2) $(1-x)^{-1} + c$
- (3) $(1-x)^{-1} - 1 + c$
- (4) None of these

19. If $\int (\cos x - \sin x) dx = \sqrt{2} \sin(x + \alpha) + c$, then $\alpha =$

- (1) $\frac{\pi}{3}$
- (2) $-\frac{\pi}{3}$
- (3) $\frac{\pi}{4}$
- (4) $-\frac{\pi}{4}$

20. $\int \frac{3x^2 - 2\sqrt{x}}{x} dx =$

- (1) $x^3 - \sqrt{x} + c$
- (2) $x^3 + \sqrt{x} + c$
- (3) $x^3 - 2\sqrt{x} + c$
- (4) $x^3 - 4\sqrt{x} + c$

21. If $\int \frac{dx}{1 + \sin x} = \tan\left(\frac{x}{2} + a\right) + b$, then

- (1) $a = \frac{\pi}{4}, b = 3$
- (2) $a = -\frac{\pi}{4}, b = 3$
- (3) $a = \frac{\pi}{4}, b = \text{arbitrary constant}$
- (4) $a = -\frac{\pi}{4}, b = \text{arbitrary constant}$

22. $\int \frac{dx}{\sin x + \cos x} =$

- (1) $\log \tan\left(\frac{\pi}{8} + \frac{x}{2}\right) + c$
- (2) $\log \tan\left(\frac{\pi}{8} - \frac{x}{2}\right) + c$
- (3) $\frac{1}{\sqrt{2}} \log \tan\left(\frac{\pi}{8} + \frac{x}{2}\right) + c$
- (4) None of these

23. $\int \frac{1}{x\sqrt{x^2-1}} dx = .$

- (1) $\cos^{-1} x + c$ (2) $\sec^{-1} x + c$
(3) $\cot^{-1} x + c$ (4) $\tan^{-1} x + c$

24. $\int \frac{\cos 2x + 2\sin^2 x}{\cos^2 x} dx =$

- (1) $2\sec x + c$ (2) $2\tan x + c$
(3) $\tan x + c$ (4) None of these

25. $\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x} dx =$

- (1) $\tan x + \cot x + c$
(2) $\tan x - \cot x + c$
(3) $\operatorname{cosec} x - \cot x + c$
(4) $\sec x - \operatorname{cosec} x + c$

26. $\int \left(\cos \frac{x}{2} - \sin \frac{x}{2} \right)^2 dx =$

- (1) $x + \cos x + c$
(2) $2\cos^2 \frac{x}{2} + c$
(3) $\frac{1}{3} \left(\cos \frac{x}{2} - \frac{x}{2} \right)^3 + c$
(4) $x - \cos x + c$

27. $\int \frac{1}{x^2} (2x+1)^3 dx =$

- (1) $4x^2 + 12x + 6\log x - \frac{1}{x} + c$
(2) $4x^2 + 12x - 6\log x - \frac{2}{x} + c$
(3) $2x^2 + 8x + 3\log x - \frac{2}{x} + c$
(4) $8x^2 + 6x + 6\log x + \frac{2}{x} + c$

28. $\int \frac{5(x^6+1)}{x^2+1} dx =$

- (1) $5(x^7+x)\tan^{-1} x + c$
(2) $x^5 - \frac{5}{3}x^3 + 5x + c$
(3) $3x^4 - 5x^2 + 15x + c$
(4) $5\tan^{-1}(x^2+1) + \log(x^2+1) + c$

29. $\int \frac{ax^{-2} + bx^{-1} + c}{x^{-3}} dx =$

- (1) $2ax^2 + 3bx^3 + 4cx^4 + k$
(2) $6ax^2 + 4bx^3 + 3cx^4 + k$
(3) $a + b + cx^2 + k$
(4) $\frac{1}{2}ax^2 + \frac{1}{3}bx^3 + \frac{1}{4}cx^4 + k$

30. $\int \frac{dx}{\sin x + \sqrt{3}\cos x} =$

- (1) $\log \tan \left(\frac{x}{2} + \frac{\pi}{2} \right) + c$
(2) $\frac{1}{2} \log \tan \left(\frac{x}{2} + \frac{\pi}{6} \right) + c$
(3) $\log \cot \left(\frac{x}{2} + \frac{\pi}{6} \right) + c$
(4) $\frac{1}{2} \log \cot \left(\frac{x}{2} + \frac{\pi}{6} \right) + c$

31. $\int \frac{x^2 + x - 6}{(x-2)(x-1)} dx =$

- (1) $x + 2\log(x-1) + c$
(2) $2x + 2\log(x-1) + c$
(3) $x + 4\log(1-x) + c$
(4) $x + 4\log(x-1) + c$

32. $\int \frac{dx}{4\cos^3 2x - 3\cos 2x} =$

- (1) $\frac{1}{3} \log[\sec 6x + \tan 6x] + c$
(2) $\frac{1}{6} \log[\sec 6x + \tan 6x] + c$
(3) $\log[\sec 6x + \tan 6x] + c$
(4) None of these

33. $\int \frac{\sin 3x}{\sin x} dx =$

- (1) $x + \sin 2x + c$ (2) $3x + \sin 2x + c$
(3) $3x + \sin^2 x + c$ (4) None of these

34. If $\int \frac{f(x) dx}{\log \sin x} = \log \log \sin x$, then $f(x) =$

- (1) $\sin x$ (2) $\cos x$
(3) $\log \sin x$ (4) $\cot x$

35. $\int \frac{\sin x + \operatorname{cosec} x}{\tan x} dx =$

- (1) $\sin x - \operatorname{cosec} x + c$
- (2) $\operatorname{cosec} x - \sin x + c$
- (3) $\log \tan x + c$
- (4) $\log \cot x + c$

36. $\int \frac{1}{\sqrt{1 + \sin x}} dx =$

- (1) $2\sqrt{2} \log \tan \left(\frac{\pi}{8} + \frac{x}{4} \right) + c$
- (2) $\frac{1}{\sqrt{2}} \log \tan \left(\frac{\pi}{8} + \frac{x}{4} \right) + c$
- (3) $\sqrt{2} \log \tan \left(\frac{\pi}{8} + \frac{x}{4} \right) + c$
- (4) $\frac{1}{2\sqrt{2}} \log \tan \left(\frac{\pi}{8} + \frac{x}{4} \right) + c$

37. $\int (\tan x - \cot x)^2 dx =$

- (1) $\tan x + \cot x + c$
- (2) $\sec x \tan x + c$
- (3) $\operatorname{cosec} x \cot x + c$
- (4) None of these

38. $\int \{1 + 2 \tan x (\tan x + \sec x)\}^{1/2} dx =$

- (1) $\log(\sec x + \tan x) + c$
- (2) $\log(\sec x + \tan x)^{1/2} + c$
- (3) $\log \sec x (\sec x + \tan x) + c$
- (4) None of these

39. $\int \frac{2x}{(2x+1)^2} dx =$

- (1) $\frac{1}{2} \log(2x+1) + \frac{1}{2(2x+1)} + c$
- (2) $\frac{1}{2} \log(2x+1) - \frac{1}{2(2x+1)} + c$
- (3) $2 \log(2x+1) + \frac{1}{2(2x+1)} + c$
- (4) $2 \log(2x+1) - \frac{1}{2(2x+1)} + c$

40. $\int \frac{\sin^2 x - \cos^2 x}{\sin^2 x \cos^2 x} dx =$

- (1) $\tan x + \cot x + c$
- (2) $\tan x + \operatorname{cosec} x + c$
- (3) $-\tan x + \cot x + c$
- (4) $\tan x + \sec x + c$

41. $\int (3 \operatorname{cosec}^2 x + 2 \sin 3x) dx =$

- (1) $3 \cot x + \frac{2}{3} \cos 3x + c$
- (2) $-\left(3 \cot x + \frac{2}{3} \cos 3x\right) + c$
- (3) $3 \cot x - \frac{2}{3} \cos 3x + c$
- (4) None of these

42. If $f'(x) = \frac{1}{x} + x$ and $f(1) = \frac{5}{2}$, then $f(x) =$

- (1) $\log x + \frac{x^2}{2} + 2$
- (2) $\log x + \frac{x^2}{2} + 1$
- (3) $\log x - \frac{x^2}{2} + 2$
- (4) $\log x - \frac{x^2}{2} + 1$

43. $\int \frac{dx}{\sqrt{1+x} + \sqrt{x}} =$

- (1) $\frac{2}{3}(1+x)^{2/3} - \frac{2}{3}x^{2/3} + c$
- (2) $\frac{3}{2}(1+x)^{2/3} + \frac{3}{2}x^{2/3} + c$
- (3) $\frac{3}{2}(1+x)^{3/2} + \frac{3}{2}x^{3/2} + c$
- (4) $\frac{2}{3}(1+x)^{3/2} - \frac{2}{3}x^{3/2} + c$

44. $\int \frac{\cos 2x - \cos 2\alpha}{\cos x - \cos \alpha} dx =$

- (1) $2[\sin x + x \cos \alpha] + c$
- (2) $2[\sin x + \sin \alpha] + c$
- (3) $2[-\sin x + x \cos \alpha] + c$
- (4) $-2[\sin x + \sin \alpha] + c$

45. $\int \frac{1 - \tan x}{1 + \tan x} dx =$

- (1) $\log \sec\left(\frac{\pi}{4} - x\right) + c$
 (2) $\log \cos\left(\frac{\pi}{4} + x\right) + c$
 (3) $\log \sin\left(\frac{\pi}{4} + x\right) + c$
 (4) None of these

46. $\int a^x da =$

- (1) $\frac{a^x}{\log_e a} + c$ (2) $a^x \log_e a + c$
 (3) $\frac{a^{x+1}}{x+1} + c$ (4) $xa^{x-1} + c$

47. The value of $\int \cot x dx$ is

- (1) $\log \cos x + c$ (2) $\log \tan x + c$
 (3) $\log \sin x + c$ (4) $\log \sec x + c$

48. The value of $\int \frac{1}{x^4} dx$ is

- (1) $\frac{1}{-3x^3} + c$ (2) $\frac{1}{3x^3} + c$
 (3) $\frac{1}{-4x^3} + c$ (4) $-\frac{1}{3x^2} + c$

49. $\int \frac{e^{5 \log x} - e^{4 \log x}}{e^{3 \log x} - e^{2 \log x}} dx =$

- (1) $e \cdot 3^{-3x} + c$ (2) $e^3 \log x + c$
 (3) $\frac{x^3}{3} + c$ (4) None of these

50. $\int \frac{x^4 + x^2 + 1}{x^2 - x + 1} dx =$

- (1) $\frac{1}{3}x^3 + \frac{1}{2}x^2 + x + c$
 (2) $\frac{1}{3}x^3 - \frac{1}{2}x^2 + x + c$
 (3) $\frac{1}{3}x^3 + \frac{1}{2}x^2 - x + c$
 (4) None of these

51. $\int \sec x dx =$

- (1) $\log \tan\left(\frac{\pi}{8} + \frac{x}{2}\right) + c$
 (2) $-\log(\sec x - \tan x) + c$
 (3) $\log(\sec x - \tan x) + c$
 (4) None of these

52. $\int \sqrt{1 + \sin x} dx =$

- (1) $\frac{1}{2}\left(\sin \frac{x}{2} + \cos \frac{x}{2}\right) + c$
 (2) $\frac{1}{2}\left(\sin \frac{x}{2} - \cos \frac{x}{2}\right) + c$
 (3) $2\sqrt{1 + \sin x} + c$
 (4) $-2\sqrt{1 - \sin x} + c$

53. $\int \operatorname{cosec}^2 x dx$ is equal to

- (1) $\cot x + c$ (2) $-\cot x + c$
 (3) $\tan^2 x + c$ (4) $-\cot^2 x + c$

54. $\int \left(2 \sin x + \frac{1}{x}\right) dx$ is equal to

- (1) $-2 \cos x + \log x + c$
 (2) $2 \cos x + \log x + c$
 (3) $-2 \sin x - \frac{1}{x^2} + c$
 (4) $-2 \cos x + \frac{1}{x^2} + c$

55. $\int \sqrt{1 + \cos x} dx$ equals

- (1) $2\sqrt{2} \sin \frac{x}{2} + c$ (2) $-2\sqrt{2} \sin \frac{x}{2} + c$
 (3) $-2\sqrt{2} \cos \frac{x}{2} + c$ (4) $2\sqrt{2} \cos \frac{x}{2} + c$

56. $\int 2 \sin x \cos x dx$ is equal to

- (1) $\cos 2x + c$ (2) $\sin 2x + c$
 (3) $\cos^2 x + c$ (4) $\sin^2 x + c$

57. $\int \tan^2 x dx$ is equal to

- (1) $\tan x + x + c$ (2) $\tan x - x + c$
 (3) $\sec x + x + c$ (4) $\sec x - x + c$

58. $\int e^{\log(\sin x)} dx =$
 (1) $\sin x + c$ (2) $-\cos x + c$
 (3) $e^{\log(\cos x)} + c$ (4) None of these

59. $\int e^{x \log a} \cdot e^x dx$ is equal to
 (1) $(ae)^x + c$ (2) $\frac{(ae)^x}{\log(ae)} + c$
 (3) $\frac{e^x}{1 + \log a} + c$ (4) None of these

60. $\int \frac{1}{\sqrt{1 + \cos x}} dx =$
 (1) $\sqrt{2} \log \left(\sec \frac{x}{2} + \tan \frac{x}{2} \right) + K$
 (2) $\frac{1}{\sqrt{2}} \log \left(\sec \frac{x}{2} + \tan \frac{x}{2} \right) + K$
 (3) $\log \left(\sec \frac{x}{2} + \tan \frac{x}{2} \right) + K$
 (4) None of these

61. $\int \frac{\cos 2x - 1}{\cos 2x + 1} dx =$
 (1) $\tan x - x + c$ (2) $x + \tan x + c$
 (3) $x - \tan x + c$ (4) $-x - \cot x + c$

62. $\int \frac{ax^3 + bx^2 + c}{x^4} dx$ equals to
 (1) $a \log x + \frac{b}{x^2} + \frac{c}{3x^3} + c$
 (2) $a \log x + \frac{b}{x} - \frac{c}{3x^3} + c$
 (3) $a \log x - \frac{b}{x} - \frac{c}{3x^3} + c$
 (4) None of these

63. The value of $\int \frac{1}{(x-5)^2} dx$ is
 (1) $\frac{1}{x-5} + c$ (2) $-\frac{1}{x-5} + c$
 (3) $\frac{2}{(x-5)^3} + c$ (d) $-2(x-5)^3 + c$

64. If $\int f(x) dx = f(x)$, then $\int [f(x)]^2 dx$ is
 (1) $\frac{1}{2} [f(x)]^2$ (2) $[f(x)]^3$
 (3) $\frac{[f(x)]^3}{3}$ (4) $[f(x)]^2$

65. If $\int \sqrt{2} \sqrt{1 + \sin x} dx = -4 \cos(ax + b) + c$ then the value of (a, b) is
 (1) $\frac{1}{2}, \frac{\pi}{4}$ (2) $1, \frac{\pi}{2}$
 (3) $1, 1$ (4) None of these

66. $\int 13^x dx$ is
 (1) $\frac{13^x}{\log 13} + c$ (2) $13^{x+1} + c$
 (3) $14x + c$ (4) $14^{x+1} + c$

67. $\int a^x dx =$
 (1) $\frac{a^x}{\log a} + c$ (2) $a^x \log a + c$
 (3) $\log a + c$ (4) $a^x + c$

68. $\int \sec x \tan x dx =$
 (1) $\sec x + \tan x + c$ (2) $\sec x + c$
 (3) $\tan x + c$ (4) $-\sec x + c$

69. $\int (\sin^4 x - \cos^4 x) dx =$
 (1) $-\frac{\cos 2x}{2} + c$ (2) $-\frac{\sin 2x}{2} + c$
 (3) $\frac{\sin 2x}{2} + c$ (4) $\frac{\cos 2x}{2} + c$

70. $\int \frac{(x+1)^2 dx}{x(x^2+1)}$ is equal to
 (1) $\log_e x + c$
 (2) $\log_e x + 2 \tan^{-1} x + c$
 (3) $\log_e \frac{1}{x^2+1} + c$
 (4) $\log_e \{x(x^2+1)\} + c$

71. The value of $\int \frac{dx}{\sqrt{1-x}}$ is

- (1) $2\sqrt{1-x} + c$ (2) $-2\sqrt{1-x} + c$
(3) $-\sin^{-1}\sqrt{x} + c$ (4) $\sin^{-1}\sqrt{x} + c$

72. $\int \frac{dx}{1-x^2} =$

- (1) $\tan^{-1}x + c$ (2) $\sin^{-1}x + c$
(3) $\frac{1}{2}\ln\left|\frac{1+x}{1-x}\right| + c$ (4) $\frac{1}{2}\ln\left|\frac{1-x}{1+x}\right| + c$

73. $\int \frac{dx}{4x^2+9} =$

- (1) $\frac{1}{2}\tan^{-1}\left(\frac{2x}{3}\right) + c$ (2) $\frac{3}{2}\tan^{-1}\left(\frac{2x}{3}\right) + c$
(3) $\frac{1}{6}\tan^{-1}\left(\frac{2x}{3}\right) + c$ (4) $\frac{1}{6}\tan^{-1}\left(\frac{3x}{2}\right) + c$

74. $\int \sqrt{1+x^2} dx =$

- (1) $\frac{x}{2}\sqrt{1+x^2} + \frac{1}{2}\log(x + \sqrt{1+x^2}) + c$
(2) $\frac{2}{3}(1+x^2)^{3/2} + c$
(3) $\frac{2}{3}x(1+x^2)^{3/2} + c$
(4) None of these

75. $\int \frac{dx}{\sqrt{x^2-a^2}}$ equals

- (1) $\sin^{-1}\left(\frac{x}{a}\right) + c$
(2) $\log_e |x + \sqrt{x^2-a^2}| + c$
(3) $\log_e |x - \sqrt{x^2-a^2}| + c$
(4) $\frac{x\sqrt{x^2-a^2}}{2+c}$

76. $\int \frac{x^2}{x^2+4} dx$ equals to

- (1) $x - 2\tan^{-1}(x/2) + c$
(2) $x + 2\tan^{-1}(x/2) + c$
(3) $x - 4\tan^{-1}(x/2) + c$
(4) $x + 4\tan^{-1}(x/2) + c$

77. $\int \sqrt{x^2+a^2} dx$ equals to

- (1) $\frac{x}{2}\sqrt{x^2+a^2} - \frac{a^2}{2}\log\{x + \sqrt{x^2+a^2}\} + c$
(2) $\frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\log\{x + \sqrt{x^2+a^2}\} + c$
(3) $\frac{x}{2}\sqrt{x^2+a^2} - \frac{a^2}{2}\log\{x - \sqrt{x^2+a^2}\} + c$
(4) $\frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\log\{x - \sqrt{x^2+a^2}\} + c$

78. $\int \frac{dx}{a^2-x^2}$ is equal to

- (1) $\frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)$ (2) $\frac{1}{2a}\sin^{-1}\left(\frac{a-x}{a+x}\right)$
(3) $\frac{1}{2a}\log\left(\frac{a+x}{a-x}\right)$ (d) $\frac{1}{2a}\log\left(\frac{a-x}{a+x}\right)$

79. $\int \frac{dx}{\sqrt{x^2+a^2}}$ is equal to

- (1) $\frac{1}{2}x\sqrt{x^2+a^2} + \frac{1}{2}a^2\log(x + \sqrt{x^2+a^2}) + c$
(2) $\frac{1}{2}\log(x^2+a^2) + c$
(3) $\log(x + \sqrt{x^2+a^2}) + c$
(4) $\log(x - \sqrt{x^2+a^2}) + c$

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (2)
2. (4)
3. (4)
4. (4)
5. (2)
6. (4)
7. (1)
8. (1)
9. (2)
10. (2)
11. (3)
12. (2)
13. (3)
14. (1, 3)
15. (3)
16. (1)
17. (1)
18. (2)
19. (3)
20. (4)
21. (4)
22. (3)
23. (2)
24. (3)
25. (4)
26. (1)
27. (1)
28. (2)
29. (4)
30. (2)
31. (4)
32. (2)
33. (1)
34. (4)
35. (1)
36. (3)
37. (4)
38. (3)
39. (1)
40. (1)

41. (2)
42. (1)
43. (4)
44. (1)
45. (3)
46. (3)
47. (3)
48. (1)
49. (3)
50. (1)
51. (2)
52. (4)
53. (2)
54. (1)
55. (1)
56. (4)
57. (2)
58. (2)
59. (2)
60. (1)
61. (3)
62. (3)
63. (2)
64. (1)
65. (1)
66. (1)
67. (1)
68. (2)
69. (2)
70. (2)
71. (2)
72. (3)
73. (3)
74. (1)
75. (2)
76. (1)
77. (2)
78. (3)
79. (3)

